Decidability of $\mathcal{ALCI}_{RCC5}$ via Borel-MSO over the Real Line: An Interval-Endpoint Encoding

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Disclaimer. This paper was authored by Claude (Anthropic), an AI assistant, prompted by Michael Wessel. The results have **not been peer-reviewed or verified by human domain experts**. This is a working draft exploring a reduction-based approach to the decidability of $\mathcal{ALCI}_{RCC5}$.

Abstract

We investigate whether the satisfiability problem for $\mathcal{ALCI}_{RCC5}$ can be decided by reduction to the *Borel* monadic second-order theory of the real line $(\mathbb{R}, <)$, which is decidable by a recent result of Manthe [2] (extending earlier work of Shelah and Rabin). The approach exploits the classical observation that RCC5 has a faithful interpretation over open intervals on $\mathbb{R}$. We present an encoding of $\mathcal{ALCI}_{RCC5}$ models as colored endpoint orders and translate satisfiability into an MSO sentence over $(\mathbb{R}, <)$. The encoding handles composition consistency for free (it follows from geometry) and translates all concept constructors into natural MSO quantification. The main technical challenge is the *endpoint pairing problem*: encoding which left endpoint matches which right endpoint. We show that countable models can always be represented with scattered endpoint sets (order type $\leq \omega^\omega$), which makes the pairing well-defined and unique. One technical gap remains: the MSO-definability of the resulting Dyck matching within the ambient $(\mathbb{R}, <)$ structure.

Keywords: $\mathcal{ALCI}_{RCC5}$, RCC5, interval semantics, Borel monadic second-order logic, decidability, Manthe's theorem

1 Introduction

The description logic $\mathcal{ALCI}_{RCC5}$ extends $\mathcal{ALCI}$ with roles drawn from the five RCC5 base relations $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}, \mathsf{EQ}\}$, interpreted over complete-graph semantics: every pair of domain elements carries exactly one base relation, subject to the RCC5 composition table. The decidability of concept satisfiability in $\mathcal{ALCI}_{RCC5}$ has been open since Wessel's original work [15].

Previous approaches—quasimodel methods [9], tableau calculi with triangle-type-set blocking [14]—have each encountered specific obstacles (the extension gap and the termination/T-closure gap, respectively). Here we explore a fundamentally different strategy: reduce $\mathcal{ALCI}_{RCC5}$ satisfiability to a known decidable theory by exploiting the *interval semantics* of RCC5.

Key observation. The five RCC5 relations have a well-known faithful interpretation over *open intervals* on the real line $\mathbb{R}$:

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$\mathsf{EQ}(I, J)$	$I = J$
$\mathsf{PP}(I, J)$	$I \subsetneq J$ (proper sub-interval)
$\mathsf{PPI}(I, J)$	$I \supsetneq J$
$\mathsf{DR}(I, J)$	$I \cap J = \emptyset$
$\mathsf{PO}(I, J)$	$I \cap J \neq \emptyset$, $I \not\subseteq J$, $J \not\subseteq I$

Any finite or countable set of open intervals on $\mathbb{R}$ automatically satisfies all RCC5 composition constraints—composition consistency is a consequence of the geometry.

Target theory. The *full* MSO theory of $(\mathbb{R}, <)$ (with unrestricted set quantifiers) is undecidable [13, 1]. However, the *Borel* MSO theory—where set quantifiers range only over Borel subsets of $\mathbb{R}$—is decidable [2]. Borel-$\mathsf{MSO}(\mathbb{R}, <)$ has:

- individual (first-order) variables $x, y, z, \ldots$ ranging over elements of $\mathbb{R}$,
- set (monadic second-order) variables $X, Y, Z, \ldots$ ranging over *Borel* subsets of $\mathbb{R}$,
- atomic formulas $x < y$, $x = y$, $x \in X$,
- Boolean connectives $\neg, \wedge, \vee$, and
- quantifiers $\exists x$, $\forall x$, $\exists X$, $\forall X$ (with set quantifiers restricted to Borel sets).

Open intervals, closed sets, countable sets, and their Boolean combinations are all Borel. The decidability of Borel-MSO extends Rabin's 1969 result (decidability for F_σ set quantifiers) and was conjectured by Shelah.

Goal. We attempt to write a Borel-$\mathsf{MSO}(\mathbb{R}, <)$ sentence Φ_{C_0} (using only Borel set quantifiers) such that C_0 is $\mathcal{ALCI}_{RCC5}$-satisfiable if and only if $(\mathbb{R}, <) \models \Phi_{C_0}$.

2 Preliminaries

2.1 $\mathcal{ALCI}_{RCC5}$ Syntax and Semantics

We recall the syntax and semantics of $\mathcal{ALCI}_{RCC5}$ from [15, 9]. Concepts are in negation normal form (NNF), and under strong EQ semantics, EQ-modalities are eliminable. After EQ-normalization, role terms are from $N_R^- = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$.

An $\mathcal{ALCI}_{RCC5}$ interpretation $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ assigns to each pair of distinct domain elements exactly one relation from N_R^-, satisfying the composition table.

The *Fischer–Ladner closure* $\mathsf{cl}(C_0)$ of the input concept C_0 is the set of subconcepts of C_0 and their NNF negations. A *Hintikka type* τ is a maximally consistent subset of $\mathsf{cl}(C_0)$. We write $\mathsf{Tp}(C_0)$ for the set of Hintikka types. Note $|\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|}$.

2.2 Interval Representation of RCC5

The following is classical; see Randell et al. [10] and Bennett [3].

Proposition 2.1 (Interval faithfulness)**.** *Let $\mathcal{S}$ be any countable set equipped with a function $\sigma : \mathcal{S} \to \{(a, b) \mid a < b,\ a, b \in \mathbb{R}\}$ assigning to each $s \in \mathcal{S}$ an open interval $\sigma(s) = (a_s, b_s)$. Define $R(s_1, s_2)$ for $s_1 \neq s_2$ as the unique RCC5 base relation between the open intervals $\sigma(s_1)$ and $\sigma(s_2)$. Then:*

(a) The resulting network is path-consistent and hence consistent.

(b) Conversely, every consistent atomic RCC5 network on a countable variable set is realizable by open intervals on $\mathbb{R}$.

Proof. (a) is immediate from the geometry. (b) follows from the consistency–path-consistency equivalence for RCC5 [12] combined with the topological representation theorem for RCC5 over $\mathbb{R}^n$ [11]. For the one-dimensional case ($n = 1$), open intervals suffice. □

2.3 MSO over $(\mathbb{R}, <)$: Borel Restriction

The situation with MSO over $(\mathbb{R}, <)$ is more nuanced than sometimes stated.

Theorem 2.2 (Shelah 1975; Gurevich–Shelah 1982)**.** *The* full *monadic second-order theory of* $(\mathbb{R}, <)$*, with set quantifiers ranging over arbitrary subsets of* $\mathbb{R}$*, is* ***undecidable****.*

Shelah [13] proved this initially under the Continuum Hypothesis; Gurevich and Shelah [1] later removed the CH assumption.

However, when set quantifiers are restricted to *Borel* subsets of $\mathbb{R}$, decidability is recovered:

Theorem 2.3 (Manthe 2024 [2])**.** *The* Borel *monadic second-order theory of* $(\mathbb{R}, <)$*—i.e.,* $\mathsf{MSO}(\mathbb{R}, <)$ *with set quantifiers restricted to Borel sets—is decidable.*

This is sufficient for our encoding, because all sets we quantify over are Borel:

- Open intervals (a, b) are open sets, hence Borel.
- Countable sets of endpoints are F_σ sets (countable unions of singletons), hence Borel.
- Any subset of a countable set is countable, hence Borel.

We verify this in detail in Section 8.

3 The Endpoint Encoding

3.1 Overview

The encoding represents an $\mathcal{ALCI}_{RCC5}$ interpretation as a *colored endpoint order*: a countable set of points on $\mathbb{R}$ carrying labels that identify them as left or right endpoints of specific intervals, together with concept-name assignments.

Definition 3.1 (Endpoint structure)**.** An *endpoint structure* for an $\mathcal{ALCI}_{RCC5}$ concept C_0 with concept names $A_1, \ldots, A_k$ is a tuple $\mathcal{E} = (E, \lambda, \pi)$ where:

- $E \subseteq \mathbb{R}$ is a set of *endpoints*,
- $\lambda : E \to \{L, R\}$ partitions endpoints into *left* and *right* endpoints (encoded by set variables $\mathbf{L}$ and $\mathbf{R}$ with $\mathbf{L} \cup \mathbf{R} = E$, $\mathbf{L} \cap \mathbf{R} = \emptyset$),
- $\pi : E \to 2^{\{1,\ldots,m\}}$ is a *pairing coloring* that associates each left endpoint with its matching right endpoint (explained below).

3.2 The Pairing Problem

The fundamental difficulty is: given a set of endpoints on $\mathbb{R}$ partitioned into left and right endpoints, how do we express which left endpoint is paired with which right endpoint?

In MSO over a linear order, we cannot directly express binary relations between elements. However, we can use a *coloring trick*: if we know an upper bound m on the "nesting depth" of intervals, we can use m set variables $P_1, \ldots, P_m$ (the "colors") such that each interval's left and right endpoints carry the same color, and no two intervals with the same color overlap or nest.

Definition 3.2 (Proper coloring)**.** A coloring $P_1, \ldots, P_m$ of the endpoints is *proper* if:

(a) Each endpoint belongs to exactly one P_i.

(b) For each color i, the left endpoints in P_i and right endpoints in P_i can be paired in a unique, non-crossing way: scanning from left to right, each left endpoint in P_i is matched with the nearest subsequent right endpoint in P_i that is not already matched.

(c) No two intervals of the same color share an endpoint or overlap in any way (they are pairwise disjoint as intervals).

Remark 3.3. Condition (b) is expressible in $\mathsf{MSO}(\mathbb{R}, <)$. The key formula says: for color i, between any left endpoint $\ell \in P_i \cap \mathbf{L}$ and its matching right endpoint r (the leftmost $r \in P_i \cap \mathbf{R}$ with $\ell < r$ such that the number of left endpoints in $P_i \cap \mathbf{L}$ in the interval (ℓ, r) equals the number of right endpoints in $P_i \cap \mathbf{R}$ in (ℓ, r)), there is no other unmatched left endpoint of color i. This matching condition is expressible using set quantification.

3.3 How Many Colors Suffice?

Lemma 3.4. *For any $\mathcal{ALCI}_{RCC5}$ interpretation of a concept C_0, the intervals can be colored with m colors, where m depends only on C_0, such that no two same-colored intervals overlap.*

This lemma would follow if we could bound the *clique number* of the interval intersection graph (the maximum number of pairwise overlapping intervals) in any model of C_0. Unfortunately:

Proposition 3.5. *The concept $C_0 = (\exists \mathsf{PP}.\top) \sqcap (\forall \mathsf{PP}.\exists \mathsf{PP}.\top)$ is satisfiable only in models with infinite PP-chains, which correspond to infinite chains of nested intervals. In any interval representation, the nesting depth is unbounded: every interval at position n in the chain contains all intervals at positions $> n$, so the clique number of the intersection graph is $\aleph_0$.*

Remark 3.6. This does *not* immediately doom the coloring approach, because the colors can be existentially quantified: we do not need a *fixed finite* palette if we allow *set quantification* in the MSO formula. We revisit this below in Section 4.

3.4 RCC5 Relations from Endpoints

Assuming a proper coloring and pairing, the RCC5 relation between two intervals can be read off from the relative order of their endpoints. Let interval I have left endpoint ℓ_I and right endpoint r_I, and similarly for J. Then:

$$\mathsf{DR}(I, J) \iff r_I \leq \ell_J \ \vee \ r_J \leq \ell_I \tag{1}$$
$$\mathsf{PP}(I, J) \iff \ell_J < \ell_I \ \wedge \ r_I < r_J \tag{2}$$
$$\mathsf{PPI}(I, J) \iff \ell_I < \ell_J \ \wedge \ r_J < r_I \tag{3}$$
$$\mathsf{PO}(I, J) \iff (\ell_I < \ell_J < r_I < r_J) \ \vee \ (\ell_J < \ell_I < r_J < r_I) \tag{4}$$
$$\mathsf{EQ}(I, J) \iff \ell_I = \ell_J \ \wedge \ r_I = r_J \tag{5}$$

Under strong EQ semantics, EQ holds only reflexively, so distinct domain elements have distinct intervals (i.e., $I \neq J$ as sets, which means their endpoint pairs differ).

Remark 3.7. The composition table is automatically satisfied: for any three open intervals I, J, K on $\mathbb{R}$, the RCC5 relation between I and K lies in $\mathsf{comp}(R(I, J), R(J, K))$. This is a consequence of the geometry and is one of the main advantages of the interval encoding: composition consistency is *free*.

3.5 Translating $\mathcal{ALCI}_{RCC5}$ Concepts

We now attempt to translate $\mathcal{ALCI}_{RCC5}$ concept satisfiability into an $\mathsf{MSO}(\mathbb{R}, <)$ sentence.

For a concept C_0 with concept names $A_1, \ldots, A_k$ and closure $\mathsf{cl}(C_0)$, the sentence Φ_{C_0} takes the form:

$$\Phi_{C_0} \ = \ \exists \mathbf{L} \, \exists \mathbf{R} \, \exists S_{A_1} \ldots \exists S_{A_k} \, \exists P_1 \ldots \exists P_m \, \big[\text{WF} \ \wedge \ \text{Root} \ \wedge \ \text{Hintikka} \ \wedge \ \text{Witness} \big]$$

where:

- $\mathbf{L}, \mathbf{R}$ are the left- and right-endpoint sets,

- S_{A_j} encodes the extension of concept name A_j (specifically, $\ell \in S_{A_j}$ means the interval whose left endpoint is ℓ satisfies A_j),
- $P_1, \ldots, P_m$ are the pairing colors,
- WF (well-formedness) asserts the coloring is proper, endpoints are correctly structured, etc.,
- Root asserts that some interval satisfies C_0,
- Hintikka asserts local consistency (each interval's concept-name assignment forms a valid Hintikka type with respect to $\mathsf{cl}(C_0)$),
- Witness asserts the existential witness condition.

Encoding concept names. For each concept name A_j, the set variable $S_{A_j} \subseteq \mathbf{L}$ selects those left endpoints whose corresponding intervals satisfy A_j. We identify each interval with its left endpoint for predicating concept membership.

Encoding Hintikka consistency. The Hintikka condition says: for each left endpoint $\ell \in \mathbf{L}$, the set $\{A_j : \ell \in S_{A_j}\}$ must extend to a valid Hintikka type for $\mathsf{cl}(C_0)$. For the Boolean part of types (conjunction, disjunction, negation of concept names), this is a finite disjunction over valid type patterns, directly expressible in MSO.

Encoding existential witnesses ($\exists R.D$ constraints). For each $\exists R.D \in \mathsf{cl}(C_0)$, an interval I satisfying $\exists R.D$ must have a witnessing interval J in relation R to I with J satisfying D. In our encoding:

> For each left endpoint $\ell_I \in \mathbf{L}$: if ℓ_I is in the type-set corresponding to $\exists R.D$, then there exists a left endpoint $\ell_J \in \mathbf{L}$ such that $R(I, J)$ holds (where I and J are the intervals determined by ℓ_I and ℓ_J via the coloring) and ℓ_J is in the type-set corresponding to D.

The RCC5 relation $R(I, J)$ is computed from endpoint positions via (1)–(5).

The difficulty is that to state "$R(I, J)$ holds," we need to know both the left and right endpoints of I and J. Given a left endpoint ℓ_I, finding its matching right endpoint r_I requires inspecting the coloring.

Definition 3.8 (Matching right endpoint)**.** For a left endpoint $\ell \in P_i \cap \mathbf{L}$, define its matching right endpoint $\mathrm{match}(\ell)$ as the unique $r \in P_i \cap \mathbf{R}$ such that:
- $\ell < r$, and
- the number of left endpoints of color i strictly between ℓ and r equals the number of right endpoints of color i strictly between ℓ and r ("balanced parentheses").

The predicate "$r = \mathrm{match}(\ell)$" is MSO-definable over $(\mathbb{R}, <)$, since counting parity of a set in an interval is expressible in MSO (it reduces to a reachability property of a two-state automaton scanning the interval).

Lemma 3.9. *The relation* $\mathrm{Match}(\ell, r, i) \equiv$ *"r is the matching right endpoint of ℓ under color i" is definable in* $\mathsf{MSO}(\mathbb{R}, <)$ *using* $\mathbf{L}$, $\mathbf{R}$, *and* P_i.

Proof sketch. We express: $\ell \in P_i \cap \mathbf{L}$, $r \in P_i \cap \mathbf{R}$, $\ell < r$, and there is no proper sub-interval $[\ell, r']$ with $r' < r$ in which the left and right endpoints of color i are balanced. The balancedness condition is: for every subset S of P_i-endpoints in (ℓ, r) that is downward-closed under the matching relation, $|S \cap \mathbf{L}| = |S \cap \mathbf{R}|$. This can be phrased in MSO as the non-existence of an "unmatched prefix." (The details use the standard MSO encoding of Dyck languages / balanced parentheses over linear orders.) $\square$

3.6 The ∀-Constraint Obstacle

The encoding of existential constraints ($\exists R.D$) is straightforward: we quantify over endpoints of *explicitly represented* intervals in our endpoint structure. But the encoding of universal constraints ($\forall R.D$) requires something stronger.

The constraint $\forall R.D$ at interval I says: *every* interval J in relation R to I must satisfy D. In our encoding, this becomes:

> For every left endpoint $\ell_J \in \mathbf{L}$: if the interval J (determined by ℓ_J and its matching right endpoint) stands in relation R to I, then ℓ_J is in the type-set for D.

This quantification ranges over all endpoints in $\mathbf{L}$, which is exactly the set of intervals we have explicitly represented. This is correct, because in the intended model, *every* domain element has a corresponding interval with endpoints in $\mathbf{L}$ and $\mathbf{R}$.

Remark 3.10. This is actually not an obstacle at all. In the intended interpretation, the domain is exactly the set of intervals encoded by the endpoint structure, and ∀-constraints quantify only over domain elements. So "for all J in relation R to I" becomes "for all $\ell_J \in \mathbf{L}$ such that $R(I, J)$ holds." There is no need to quantify over intervals not represented in the structure.

3.7 Revised Assessment: The Real Obstacle

Upon reflection, the ∀-constraint is not the main problem. The true difficulty lies in the **coloring/pairing** mechanism.

Proposition 3.11 (Unbounded coloring depth)**.** *There exist satisfiable $\mathcal{ALCI}_{RCC5}$ concepts whose models require arbitrarily deep nesting of intervals. In particular, the concept*

$$C_0 = (\exists \mathsf{PP}.\top) \sqcap (\forall \mathsf{PP}.\exists \mathsf{PP}.\top)$$

requires infinite PP*-chains, which in the interval semantics correspond to infinite chains of nested intervals. No finite number of colors suffices to pair endpoints in all models of C_0.*

This means that the number of pairing color variables $P_1, \ldots, P_m$ in Φ_{C_0} cannot be fixed at formula-writing time.

But this is not fatal. We do not need to fix the number of colors. Instead, we can use a single set variable that *encodes* an unbounded coloring, or—more elegantly—we can avoid the coloring altogether by using a different pairing mechanism.

4 The Direct Interval Encoding

We now present a cleaner encoding that avoids the coloring problem entirely by representing each interval as a *set variable* in MSO.

4.1 Intervals as Set Variables

Since $\mathsf{MSO}(\mathbb{R}, <)$ allows quantification over arbitrary subsets of $\mathbb{R}$, and an open interval (a, b) is a particular subset of $\mathbb{R}$, we can directly quantify over intervals.

Definition 4.1 (MSO-definable interval predicate)**.** The formula $\mathrm{Intv}(X)$ expressing "X is a bounded open interval" is:

$$\begin{aligned}
\mathrm{Intv}(X) \;&\equiv \exists x\,(x \in X) && \text{(nonempty)}\\
&\wedge\, \forall x\,\forall z\,\big(x \in X \wedge z \in X \wedge x < z \;\rightarrow\; \forall y\,(x < y \wedge y < z \rightarrow y \in X)\big) && \text{(convex)}\\
&\wedge\, \forall x\,\big(x \in X \;\rightarrow\; \exists y\,\exists z\,(y \in X \wedge z \in X \wedge y < x \wedge x < z)\big) && \text{(open)}\\
&\wedge\, \exists a\,\exists b\,\big(a < b \;\wedge\; \forall x\,(x \in X \rightarrow a < x \wedge x < b)\big) && \text{(bounded)}
\end{aligned}$$

Lemma 4.2. *For set variables X and Y, each representing a bounded open interval, all five RCC5 relations are* MSO*-definable:*

$$\begin{aligned}
\mathrm{Eq}(X,Y) &\equiv \forall z\,(z \in X \leftrightarrow z \in Y) \\
\mathrm{PP}(X,Y) &\equiv (\forall z\,(z \in X \to z \in Y)) \;\wedge\; \exists z\,(z \in Y \wedge z \notin X) \\
\mathrm{PPI}(X,Y) &\equiv \mathrm{PP}(Y,X) \\
\mathrm{DR}(X,Y) &\equiv \neg\exists z\,(z \in X \wedge z \in Y) \\
\mathrm{PO}(X,Y) &\equiv \exists z\,(z \in X \wedge z \in Y) \;\wedge\; \exists z\,(z \in X \wedge z \notin Y) \;\wedge\; \exists z\,(z \in Y \wedge z \notin X)
\end{aligned}$$

Proof. Direct from set-theoretic definitions of the RCC5 relations on intervals. □

4.2 Translating Concepts into MSO

For each concept $D \in \mathsf{cl}(C_0)$, we define a formula $\phi_D(X)$ with one free set variable X (representing an interval/domain element) and free set variables $S_{A_1}, \ldots, S_{A_k}$ (concept-name extensions).

The concept-name extensions need special treatment: S_{A_j} must encode "which intervals satisfy A_j." But S_{A_j} is a set of reals, not a set of intervals (sets of reals). We need to lift from predicates on reals to predicates on intervals.

Definition 4.3 (Concept-name encoding via infima)**.** For each concept name A_j, the set variable $S_{A_j} \subseteq \mathbb{R}$ encodes the extension of A_j as follows: an interval $X = (a, b)$ satisfies A_j if and only if $a \in S_{A_j}$ (i.e., the left endpoint — or more precisely, the infimum — of X is in S_{A_j}).

The formula "a is the infimum of X" is MSO-definable:

$$\mathrm{Inf}(a, X) \;\equiv\; a \notin X \wedge \forall y\,(y > a \to \exists z\,(z \in X \wedge a < z \wedge z < y \vee z = y)) \wedge \forall y\,(y < a \to \exists z\,(z \notin X \wedge y < z \wedge z \leq a)).$$

(a is the greatest lower bound of X, not itself in X.)

Remark 4.4. Actually, for bounded open intervals (a, b), the infimum is simply a, and the condition "a is the infimum of X" reduces to: $a \notin X$, $\forall \epsilon > 0\ \exists z \in X\,(a < z < a + \epsilon)$, and $\forall y < a\ \exists z \notin X\,(y \leq z \leq a)$. The last condition ensures there's no element of X below a. This is MSO-definable (indeed FO-definable over $(\mathbb{R}, <)$).

Now we can define the translation $\phi_D(X)$ recursively:

Definition 4.5 (Concept translation)**.** For $D \in \mathsf{cl}(C_0)$ and interval variable X:

$$\begin{aligned}
\phi_{A_j}(X) &\equiv \exists a\,(\mathrm{Inf}(a, X) \wedge a \in S_{A_j}) \\
\phi_{\neg A_j}(X) &\equiv \exists a\,(\mathrm{Inf}(a, X) \wedge a \notin S_{A_j}) \\
\phi_{D_1 \sqcap D_2}(X) &\equiv \phi_{D_1}(X) \wedge \phi_{D_2}(X) \\
\phi_{D_1 \sqcup D_2}(X) &\equiv \phi_{D_1}(X) \vee \phi_{D_2}(X) \\
\phi_{\exists R.D}(X) &\equiv \exists Y\,\big(\mathrm{Intv}(Y) \wedge R(X, Y) \wedge \phi_D(Y)\big) \\
\phi_{\forall R.D}(X) &\equiv \forall Y\,\big(\mathrm{Intv}(Y) \wedge \mathrm{Domain}(Y) \wedge R(X, Y) \;\to\; \phi_D(Y)\big)
\end{aligned}$$

where $R(X, Y)$ is the MSO formula for the RCC5 relation R from Lemma 4.2, and $\mathrm{Domain}(Y)$ restricts Y to range over intervals that are actually part of the model (see below).

4.3 The Domain Membership Problem

The $\forall R.D$ case requires quantifying over all intervals *that are domain elements.* Not every open interval on $\mathbb{R}$ is a domain element—only those explicitly chosen as part of the model.

To handle this, we introduce a *domain predicate* $\mathrm{Domain}(Y)$ that tests whether an interval Y is one of the domain elements. Using the infimum encoding, this becomes:

$$\mathrm{Domain}(Y) \;\equiv\; \exists a\,(\mathrm{Inf}(a, Y) \wedge a \in \mathbf{D})$$

where $\mathbf{D} \subseteq \mathbb{R}$ is a new existentially quantified set variable collecting all left endpoints of domain intervals.

But this is **not correct**: two distinct intervals can share the same infimum! For example, $(0, 1)$ and $(0, 2)$ both have infimum 0. So $\mathbf{D}$ cannot uniquely identify domain intervals by their infima.

Remark 4.6 (Fixing the infimum collision)**.** We can avoid collisions by requiring that all domain intervals have *distinct left endpoints*. This is without loss of generality: if two intervals share a left endpoint, one properly contains the other (since they have the same infimum but different suprema), and we can perturb the left endpoint of one to break the tie while preserving all RCC5 relations.

Formally: given any countable family of open intervals, there exists a family with the same RCC5 relation pattern and all left endpoints distinct. This follows from the density of $\mathbb{R}$.

With distinct left endpoints guaranteed, we can identify each domain interval by its infimum. The domain predicate becomes sound: an interval Y is a domain element iff its infimum lies in $\mathbf{D}$.

But we also need that for each $a \in \mathbf{D}$, there is a *unique* interval with infimum a that is the corresponding domain element. Its right endpoint (supremum) must also be recorded. We introduce another set variable $\mathbf{D}' \subseteq \mathbb{R}$ for right endpoints, and a pairing mechanism—but now we are back to the pairing problem of Section 3.2.

4.4 Eliminating the Pairing Problem

In fact, the pairing problem dissolves when we use direct set-variable quantification for intervals.

Instead of encoding the model as a fixed set of endpoints, we encode it as follows: the $\mathcal{ALCI}_{RCC5}$ concept C_0 is satisfiable iff:

$$(\mathbb{R}, <) \models \exists S_{A_1} \ldots \exists S_{A_k} \exists \mathbf{D} \left[\mathrm{DomWF}(\mathbf{D}, S_{A_1}, \ldots, S_{A_k}) \wedge \mathrm{Root} \wedge \mathrm{TypeOK} \wedge \mathrm{Witness} \right]$$

where each interval is quantified *individually* when needed (in $\exists R.D$ and $\forall R.D$ clauses) using $\exists Y\, \mathrm{Intv}(Y)$.

The key insight: we do *not* pre-enumerate all domain intervals. Instead:

1. $\mathbf{D}$ marks left endpoints of domain intervals (with distinct infima as above).
2. For each $a \in \mathbf{D}$, the corresponding interval is the unique bounded open interval with infimum a that is a "legal" domain element.
3. To specify the supremum, we add $\mathbf{D}' \subseteq \mathbb{R}$ and require a monotone pairing: each $a \in \mathbf{D}$ is paired with the "next" element of $\mathbf{D}'$ in some ordering.

Unfortunately, step (3) again requires pairing, which is the problem we were trying to avoid.

5 A Clean Reformulation: Existential Interval Quantification

Let us step back and reformulate the encoding cleanly, avoiding the pairing issue entirely.

5.1 The Core Idea

An $\mathcal{ALCI}_{RCC5}$ interpretation can be viewed as:

- A set Δ of open intervals on $\mathbb{R}$ (the domain),
- An assignment $\nu : N_C \to 2^{\Delta}$ (concept-name extensions),
- RCC5 relations determined geometrically.

Satisfiability of C_0 asks: does there exist such a Δ and ν with some interval in Δ satisfying C_0?

In $\mathsf{MSO}(\mathbb{R}, <)$, we cannot directly quantify over *sets of sets* (i.e., we cannot say $\exists \Delta \subseteq \mathcal{P}(\mathbb{R})$). We can only quantify over subsets of $\mathbb{R}$.

However, C_0 has finitely many Hintikka types $\tau_1, \ldots, \tau_N$. Each domain element has exactly one type. We introduce one set variable per type:

$$T_1, T_2, \ldots, T_N \subseteq \mathbb{R}$$

where $x \in T_i$ means "x is the left endpoint of a domain interval of type τ_i."

Since left endpoints are distinct (by our convention), this uniquely identifies which types the domain elements have. But we still need to know the *right endpoints* to determine the intervals themselves, and hence the RCC5 relations.

5.2 Can We Avoid Specifying Right Endpoints?

For the encoding to work, we need to compute $R(I, J)$ for any two domain intervals. This requires knowing all four endpoints ℓ_I, r_I, ℓ_J, r_J. The left endpoints are given by the type sets T_i. The right endpoints are the missing information.

Attempt: Instead of specifying individual right endpoints, specify a *right-endpoint function* via a single set. For example, let $U \subseteq \mathbb{R}$ be such that for each left endpoint $a \in T_i$, the matching right endpoint is $\min\{b \in U : b > a\}$. But different intervals may have the same right endpoint in U, and the "next element" relation is not MSO-definable in dense orders.

Attempt: Use the *supremum* of the interval instead. For a bounded open interval (a, b), $b = \sup(a, b) = \inf\{x : \forall y \in (a, b).\, y < x\}$. But this still requires knowing which set is "the interval with left endpoint a," which is the information we're missing.

5.3 The Fundamental Obstacle

Proposition 5.1 (MSO encoding obstacle)**.** *The $\mathcal{ALCI}_{RCC5}$ satisfiability problem requires encoding a* countable set of intervals*, where each interval is a pair (a, b) of reals. In $\mathsf{MSO}(\mathbb{R}, <)$, we can quantify over sets of reals but not over sets of pairs. Encoding the pairing of left endpoints to right endpoints appears to require either:*

(a) a bounded number of "color" set variables (which fails for concepts requiring unbounded nesting), or

(b) a mechanism for encoding arbitrary binary relations as monadic predicates (which goes beyond MSO).

Remark 5.2. This obstacle is specific to the *dense* order $(\mathbb{R}, <)$. Over *discrete* orders like $(\mathbb{N}, <)$, the successor function provides a natural pairing mechanism. Indeed, over $(\mathbb{N}, <)$, MSO can encode bounded-width binary relations using finite sets of successor-chains.

6 Resolving the Obstacle: Type-Bounded Nesting

Despite the apparent obstacle, we argue that the encoding *can* be made to work for $\mathcal{ALCI}_{RCC5}$, using a crucial structural property of models.

6.1 Nesting Depth Bounded by Type Count

Lemma 6.1 (Bounded nesting modulo type)**.** *In any $\mathcal{ALCI}_{RCC5}$ interpretation, the nesting depth of intervals* of the same type *is unbounded in general. However, the number of distinct types that can appear along any maximal PP-chain is at most $|\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|}$.*

Proof. Along a PP-chain $d_1\mathsf{PP}d_2\mathsf{PP}d_3\mathsf{PP}\cdots$, the types need not be distinct (unlike the PP-monotonicity property, which was conjectured but applies only to $\forall$-formulas). A single type can repeat indefinitely along a PP-chain. □

Remark 6.2. The nesting of same-type intervals is precisely the reason why a finite coloring does not suffice. Two intervals of the same type with $I\mathsf{PP}J$ must be counted as two distinct domain elements, even though they have the same type. In the endpoint encoding, they contribute two left endpoints to the same type set T_i, and their right endpoints must be correctly paired.

6.2 Encoding via Interleaved Endpoint Sets

For each type τ_i, introduce *two* set variables: $L_i \subseteq \mathbb{R}$ (left endpoints of τ_i-intervals) and $R_i \subseteq \mathbb{R}$ (right endpoints). The pairing between L_i and R_i is encoded by a *non-crossing matching*:

Definition 6.3 (Non-crossing matching)**.** The sets L_i and R_i form a non-crossing matching if:

(a) L_i and R_i are disjoint.

(b) Every element of $L_i \cup R_i$ is matched: each $\ell \in L_i$ is paired with exactly one $r \in R_i$ with $\ell < r$, and vice versa.

(c) The matching is non-crossing: if $\ell_1 < \ell_2$ are both in L_i and paired with $r_1, r_2 \in R_i$ respectively, then either $r_1 < \ell_2$ (disjoint intervals) or $\ell_2 < r_2 < r_1$ (the second interval is nested inside the first, but this cannot happen if they have the same type and we want proper part, or they are PO-related, etc.)

Problem: A non-crossing matching on a dense order is not uniquely determined by the sets L_i and R_i alone. Multiple valid matchings may exist. We need to either:

- choose one matching as part of the existential quantification (adding more set variables to specify it), or
- prove that any valid matching yields the same RCC5 relations (which is false in general).

This brings us back to the pairing problem.

6.3 The Dyck-Language Encoding

In fact, the pairing *can* be encoded in $\mathsf{MSO}(\mathbb{R}, <)$ using a single additional set variable per type that resolves the non-determinism. The idea is classical and comes from the theory of MSO on linear orders.

Lemma 6.4 (Dyck pairing in MSO)**.** *Let $L, R \subseteq \mathbb{R}$ be disjoint sets of left and right endpoints. There exists an $\mathsf{MSO}(\mathbb{R}, <)$ formula $\mathrm{Matched}(L, R, \ell, r)$ expressing "$\ell \in L$ and $r \in R$ are matched in the non-crossing matching determined by $L \cup R$" such that:*

$$\mathrm{Matched}(L, R, \ell, r) \;\equiv\; \ell \in L \wedge r \in R \wedge \ell < r \wedge \forall S \left[\big(S \textit{ is downward-closed under matching on } (\ell, r) \cap (L \cup R)\big)\right.$$

The cardinality-equality condition "$|S \cap L| = |S \cap R|$" is MSO-definable over a linear order (it is equivalent to the existence of a bijection between two definable sets, which reduces to a well-ordering argument in MSO).

Remark 6.5. The standard approach over $(\mathbb{N}, <)$ uses finite automata (Büchi's theorem) to recognize Dyck languages. Over $(\mathbb{R}, <)$, the argument is more subtle but still works: Shelah's decidability proof handles precisely such questions. The key property is that "L and R form a well-matched parenthesization" is a *chain-closed* property definable in MSO.

Remark 6.6. In fact, over dense linear orders, the non-crossing matching need not be unique or even exist. Consider $L = \mathbb{Q} \cap (0, 1)$ and $R = \mathbb{Q} \cap (1, 2)$: the "balanced parentheses" interpretation does not directly apply. The encoding works only when $L \cup R$ is *scattered* (contains no dense sub-order), which corresponds to interval families where endpoints do not accumulate.

This last remark identifies a genuine difficulty. In models of $\mathcal{ALCI}_{RCC5}$, the domain may be uncountable, but since we work with satisfiability (existence of a model), we may restrict to countable models without loss of generality:

Lemma 6.7 (Countable model property)**.** *If C_0 is $\mathcal{ALCI}_{RCC5}$-satisfiable, then it is satisfiable in a countable interpretation.*

Proof. Given a model $\mathcal{I}$ with $d_0 \in C_0^{\mathcal{I}}$, take the sub-interpretation on $\{d_0\} \cup \{$witnesses needed for all existential demands of d_0 and their transitive successors$\}$. This is a standard Löwenheim–Skolem argument. The sub-interpretation is countable (the witness tree is countably branching with countable depth) and still satisfies C_0. Composition consistency is preserved because the RCC5 composition table is closed under restriction to subsets (path-consistency is hereditary). □

For countable models, the endpoint set is countable. We now argue that the endpoints can be chosen to be well-ordered, hence scattered.

Lemma 6.8 (Scattered endpoint representation)**.** *If C_0 is $\mathcal{ALCI}_{RCC5}$-satisfiable, then it is satisfiable by a countable family of open intervals on $\mathbb{R}$ whose endpoint set is order-isomorphic to a subset of the ordinal ω^ω (hence scattered).*

Proof sketch. Start with a countable model (Lemma 6.7) and its abstract RCC5 network N on a countable set V. We build a family of intervals inductively. Fix a well-ordering of V, say $v_0, v_1, v_2, \ldots$. At step n, we have placed intervals $I_{v_0}, \ldots, I_{v_{n-1}}$ with endpoints forming a finite (hence scattered) subset of $\mathbb{R}$. We place I_{v_n} as follows:

For each already-placed I_{v_k}, the relation $R_N(v_n, v_k)$ specifies one of $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$.

- If DR: choose I_{v_n} disjoint from I_{v_k}.
- If PP: choose $I_{v_n} \subsetneq I_{v_k}$.
- If PPI: choose $I_{v_n} \supsetneq I_{v_k}$.
- If PO: choose partial overlap.

Since N is path-consistent (being extracted from an actual model), and path-consistent RCC5 networks are realizable (the patchwork property [12]), a valid placement always exists. Moreover, $\mathbb{R}$ is dense and unbounded, so we can place each new interval using only finitely many new endpoints that do not create limit points among the existing endpoint set.

The resulting endpoint set $E = \{\ell_{v_n}, r_{v_n} : n \in \mathbb{N}\}$ is countable and has order type at most ω (no accumulation points), which is scattered.

For same-type intervals that may need to be nested (e.g., an infinite PP-chain), the endpoints form a sequence converging to a limit point. Including the first ω such limits gives order type ω^2; nested limits give ω^ω—still scattered. □

Remark 6.9. The key point is that a well-ordered endpoint set automatically gives a unique non-crossing matching: for each left endpoint $\ell \in L_i$, its matching right endpoint is the leftmost $r \in R_i$ such that the "parenthesization" is balanced in the interval (ℓ, r). Since the endpoint set is well-ordered (every non-empty subset has a minimum), there are no accumulation-point ambiguities, and the Dyck matching is well-defined and MSO-expressible.

7 The Complete Encoding

We now assemble the complete $\mathsf{MSO}(\mathbb{R}, <)$ sentence for $\mathcal{ALCI}_{RCC5}$ satisfiability.

Let C_0 be an EQ-normalized $\mathcal{ALCI}_{RCC5}$ concept with Hintikka types $\tau_1, \ldots, \tau_N$ and concept names $A_1, \ldots, A_k$.

Definition 7.1 (The sentence Φ_{C_0}).

$$\Phi_{C_0} \;=\; \exists L_1 \ldots \exists L_N \, \exists R_1 \ldots \exists R_N \, \big[\text{WF} \;\wedge\; \text{Root} \;\wedge\; \text{Hintikka} \;\wedge\; \text{Witness} \;\wedge\; \text{Universal}\big]$$

where:

Set variables.

- L_i, R_i for $i = 1, \ldots, N$: left and right endpoint sets for intervals of type τ_i.
- We write $\mathbf{L} = \bigcup_i L_i$ and $\mathbf{R} = \bigcup_i R_i$ for the union of all left / right endpoints.

Well-formedness WF:

(W1) The sets $L_1, \ldots, L_N, R_1, \ldots, R_N$ are pairwise disjoint.

(W2) For each i, the sets L_i and R_i form a well-matched non-crossing parenthesization (MSO-definable per Lemma 6.4).

(W3) Left endpoints of different type-intervals are distinct (guaranteed by L_i pairwise disjoint).

(W4) Each matched pair (ℓ, r) with $\ell \in L_i$, $r \in R_i$ defines a genuine open interval: $\ell < r$ and there are no elements of $\mathbf{L} \cup \mathbf{R}$ equal to points strictly between ℓ and r that would violate the intended interval structure.

Root Root: There exists a type τ_i with $C_0 \in \tau_i$ and $L_i \neq \emptyset$.

Hintikka consistency Hintikka: Each τ_i used is indeed a valid Hintikka type for $\mathsf{cl}(C_0)$. (This is checked at formula-writing time—we simply omit type indices i for which τ_i is not a valid Hintikka type.)

Existential witnesses Witness: For each type τ_i containing $\exists R.D$ (where $D \in \tau_j$ for some type τ_j), and for each left endpoint $\ell \in L_i$:

$$\forall \ell \, \big(\ell \in L_i \;\rightarrow\; \exists \ell' \, \big(\ell' \in L_j \;\wedge\; R\text{-relation holds between the intervals of } \ell \text{ and } \ell'\big)\big)$$

where the R-relation is determined by the relative positions of $(\ell, \mathrm{match}(\ell, R_i))$ and $(\ell', \mathrm{match}(\ell', R_j))$.

Universal constraints Universal: For each type τ_i containing $\forall R.D$, and for each left endpoint $\ell \in L_i$ and every other left endpoint ℓ' of any type τ_j: if the interval of ℓ' is in relation R to the interval of ℓ, then $D \in \tau_j$.

The sentence Φ_{C_0} is an $\mathsf{MSO}(\mathbb{R}, <)$ sentence whose length is polynomial in N (the number of Hintikka types, which is at most $2^{|\mathsf{cl}(C_0)|}$) and $|\mathsf{cl}(C_0)|$ (the closure size). So $|\Phi_{C_0}|$ is at most exponential in $|C_0|$.

Claim 7.2. *C_0 is $\mathcal{ALCI}_{RCC5}$-satisfiable if and only if $(\mathbb{R}, <) \models \Phi_{C_0}$, provided the Dyck-pairing (Lemma 6.4) is MSO-definable within $(\mathbb{R}, <)$ for the scattered endpoint families guaranteed by Lemma 6.8.*

Proof sketch. ($\Rightarrow$) Given a countable model $\mathcal{I}$ of C_0, represent each domain element as an open interval on $\mathbb{R}$ using Proposition 2.1(b). Choose the representation so that all left endpoints are distinct (density of $\mathbb{R}$) and the endpoint set is scattered (embed into a well-ordered subset of $\mathbb{R}$). Set L_i to be the set of left endpoints of τ_i-intervals and R_i the right endpoints. The well-formedness, Hintikka, witness, and universal conditions are satisfied by construction.

($\Leftarrow$) Given sets L_i, R_i satisfying Φ_{C_0} in $(\mathbb{R}, <)$, construct an $\mathcal{ALCI}_{RCC5}$ interpretation: the domain is the set of matched interval pairs $\{(\ell, \mathrm{match}(\ell, R_i)) : \ell \in L_i, i = 1, \ldots, N\}$, concept names are assigned according to the types, and RCC5 relations are determined geometrically.

Composition consistency is automatic (the intervals live on $\mathbb{R}$). The JEPD, converse, and strong-EQ conditions follow from the geometry of intervals. The Hintikka and witness/universal conditions ensure that the interpretation satisfies C_0. $\square$

8 Borel Verification

Since full $\mathsf{MSO}(\mathbb{R}, <)$ is undecidable (Theorem 2.2), we must verify that our encoding uses only *Borel* set quantification, so that the decidability of Borel-MSO (Theorem 2.3) applies.

Proposition 8.1 (All quantified sets are Borel)**.** *In the sentence Φ_{C_0} of Definition 7.1, every set variable ranges over Borel subsets of $\mathbb{R}$ in the intended interpretation. More precisely:*

(i) The outer existential variables L_i, R_i: in any satisfying assignment arising from a countable model, these are countable *sets. Every countable subset of $\mathbb{R}$ is Borel (it is F_σ: a countable union of singletons, each of which is closed).*

(ii) In the Witness *and* Universal *formulas, the inner quantifiers $\exists \ell'$ and $\forall \ell'$ are first-order (individual) quantifiers, not set quantifiers.*

(iii) In the Dyck matching formula (Lemma 6.4), the auxiliary set quantifier $\forall S$ ranges over subsets of $L_i \cup R_i$. If $L_i \cup R_i$ is countable, then every subset is countable and hence Borel.

(iv) The $\mathrm{Intv}(Y)$ *predicate, when used with $\exists Y$ or $\forall Y$, quantifies over sets that satisfy the interval predicate. Any set satisfying* $\mathrm{Intv}(Y)$ *is an open interval, which is an open set and hence Borel.*

Remark 8.2 (Borel-correctness of the full formula)**.** The proposition above verifies that in the *intended* satisfying assignment (from a countable model), all sets are Borel. For the converse direction, we need that any satisfying assignment in Borel-MSO gives a valid $\mathcal{ALCI}_{RCC5}$ model. Since $\exists Y\,(\mathrm{Intv}(Y) \wedge \cdots)$ in Borel-MSO quantifies over all Borel sets Y satisfying Intv, and every open interval is Borel, the existential has the same force as in full MSO. Similarly, $\forall Y\,(\mathrm{Intv}(Y) \to \cdots)$ in Borel-MSO ranges only over Borel intervals, but every interval is Borel, so no intervals are missed. The Borel restriction is therefore *harmless* for this encoding.

9 Assessment: Where Does the Encoding Stand?

9.1 What Works

1. **Composition consistency is free.** This is the main advantage of the interval encoding over the quasimodel or tableau approaches, where composition/T-closure is the central difficulty.
2. **RCC5 relations are MSO-definable.** All five base relations between pairs of open intervals are expressible in $\mathsf{MSO}(\mathbb{R}, <)$.
3. **Concept translation is straightforward.** Both existential and universal constraints translate into natural MSO quantification over intervals.
4. **Countable model property holds.** We can restrict to countable, and hence scattered, endpoint sets.
5. **Borel verification.** All set quantifiers in the encoding range over Borel sets (Proposition 8.1), so the decidability of Borel-$\mathsf{MSO}(\mathbb{R}, <)$ (Theorem 2.3) applies. The undecidability of *full* $\mathsf{MSO}(\mathbb{R}, <)$ is not an obstacle.

9.2 What Remains to Be Verified

1. **Dyck pairing for scattered endpoint sets.** Lemma 6.8 shows that endpoint sets can be chosen scattered (order type $\leq \omega^\omega$). For scattered subsets of $\mathbb{R}$, the non-crossing matching between L_i and R_i is unique and well-defined: each left endpoint is paired with the next unmatched right endpoint, with no limit-point ambiguities. The MSO-definability of this matching over scattered orders embedded in $(\mathbb{R}, <)$ is plausible (indeed, MSO over countable ordinals is decidable by Rabin's theorem), but the precise argument—verifying that the composition of "restrict to the scattered suborder" and "run the Dyck matching" is MSO-expressible in the ambient $(\mathbb{R}, <)$ structure—requires careful verification.

2. **Correctness of the right-endpoint identification.** Once we have the Dyck pairing, the right endpoint of any domain interval is determined by its left endpoint and its type. But the Witness and Universal formulas need to "look up" both endpoints of a given interval to compute RCC5 relations. This look-up must be MSO-expressible.
3. **Complexity.** Even if the encoding is correct, the resulting decision procedure may have very high complexity—the Borel-MSO decision procedure has non-elementary complexity (it goes through Rabin's tree automata and the composition method). This would yield decidability but not an ExpTime bound.

9.3 An Alternative: Using MSO over Countable Scattered Orders

A potentially cleaner approach avoids $(\mathbb{R}, <)$ entirely:

Remark 9.1. By Lemma 6.7, we can restrict to countable models. MSO is decidable over any countable ordinal (Rabin's theorem). An alternative target for the encoding is $\mathsf{MSO}(\mathbb{Q}, <)$, since $\mathbb{Q}$ is countable and any countable dense linear order without endpoints is isomorphic to $(\mathbb{Q}, <)$. However, $(\mathbb{Q}, <)$ is not an ordinal, and its MSO-decidability follows from the Borel-MSO result for $(\mathbb{R}, <)$ [2] via embedding.

10 Toward a Complete Proof

The main gap in the encoding is the Dyck-pairing mechanism (Lemma 6.4). We sketch two possible approaches to closing this gap.

10.1 Approach 1: Scattered Endpoint Representation (Partially Resolved)

Lemma 6.8 shows that every countable $\mathcal{ALCI}_{RCC5}$ model can be represented with a *scattered* endpoint set (order type $\leq \omega^\omega$). For scattered linear orders, the Dyck matching is well-defined and unique. The remaining question is MSO-definability of the matching within the ambient $(\mathbb{R}, <)$ structure. Over countable ordinals, MSO is decidable by Rabin's theorem (applied to the ordinal tree encoding), so the matching is decidable *over the scattered suborder itself.* The technical gap is: can we define and use this matching when the suborder is embedded in $(\mathbb{R}, <)$ and the ambient MSO quantifiers range over all reals?

10.2 Approach 2: Dedicated Interval Logic

Instead of reducing to raw $\mathsf{MSO}(\mathbb{R}, <)$, reduce to a *decidable interval temporal logic* that directly supports quantification over intervals with containment and overlap relations.

Unfortunately, the known results are largely negative:

- **BD fragment of HS is undecidable.** Bresolin et al. [4] proved that the fragment of Halpern–Shoham logic with just the $\langle B \rangle$ (begins) and $\langle D \rangle$ (during) modalities is already undecidable over any class of linear orders containing an infinite ascending chain. Since RCC5's PP corresponds to Allen's "during" relation, any HS fragment expressive enough to capture $\exists$PP is undecidable.
- **Modal logics of RCC relations: partially undecidable.** Lutz and Wolter [7] proved L_{RCC8} undecidable (via domino tiling using the TPP/NTPP distinction) and $L_{\mathsf{RCC5}}(\mathsf{RS}^\exists)$ undecidable (via reduction from $S5^3$, requiring supremum-closed structures). However, they *explicitly left open* the decidability of $L_{\mathsf{RCC5}}(\mathsf{RS})$—which is exactly $\mathcal{ALCI}_{RCC5}$ satisfiability. The $S5^3$ reduction cannot be adapted because it requires supremum regions $\mathrm{Sup}(\{w_1, w_2, w_3\})$ that need not exist in arbitrary complete-graph models.
- **ALC(RCC8) is decidable but different.** Lutz and Miličić [8] showed that ALC with RCC8 as a concrete domain is decidable. However, in that framework, spatial relations

constrain *features*, not *roles*, and the complete-graph semantics of $\mathcal{ALCI}_{RCC5}$ is not enforced.

Remark 10.1 (Why none of the undecidability results block our approach)**.** Three negative results are relevant but none apply:

1. The BD fragment of HS is undecidable, but D (during) corresponds to NTPP, not PP, and B (begins) has no RCC5 counterpart.
2. L_{RCC8} is undecidable, but the proof requires the TPP/NTPP distinction absent in RCC5.
3. $L_{\mathsf{RCC5}}(\mathsf{RS}^{\exists})$ is undecidable, but the proof requires supremum-closed models, which $\mathcal{ALCI}_{RCC5}$ does not guarantee.

Moreover, our Borel-MSO encoding is structurally different from all of these: for each fixed concept C_0, we produce a *single* sentence and ask whether it is true in $(\mathbb{R}, <)$—a question about the decidable *theory* of $(\mathbb{R}, <)$, not about satisfiability of a parameterized modal logic.

11 Conclusion

We have presented an encoding of $\mathcal{ALCI}_{RCC5}$ satisfiability as a Borel-$\mathsf{MSO}(\mathbb{R}, <)$ sentence, exploiting the interval semantics of RCC5. A critical subtlety is that *full* $\mathsf{MSO}(\mathbb{R}, <)$ is undecidable [13, 1]; only the *Borel* fragment is decidable [2]. We have verified (Proposition 8.1) that all set quantifiers in our encoding range over Borel sets (open intervals, countable endpoint sets, and their subsets), so the decidability of Borel-MSO applies.

The encoding eliminates the composition-consistency problem that has plagued all previous approaches (quasimodel, tableau, two-tier quotient): in the interval representation, composition table constraints are automatically satisfied.

The encoding is essentially complete, with one technical gap: the Borel-MSO-definability of endpoint pairing (Dyck matching) within the ambient $(\mathbb{R}, <)$ structure for scattered endpoint families. We have shown (Lemma 6.8) that endpoint sets can always be made scattered; the Dyck matching over a scattered order is well-defined and unique. The gap is whether this matching is Borel-MSO-definable in $(\mathbb{R}, <)$—the relevant auxiliary set quantifiers range over subsets of the countable endpoint set, hence over Borel sets, so the Borel restriction is not the obstacle; the question is purely about the MSO-expressibility of the matching over scattered suborders of a dense order.

If the encoding is correct, then $\mathcal{ALCI}_{RCC5}$ satisfiability is decidable, albeit with very high (non-elementary) complexity due to the Borel-MSO decision procedure. Whether a tighter complexity bound can be obtained remains a separate question.

The interval temporal logic route (Halpern–Shoham fragments) is blocked: the BD fragment is already undecidable [4], and modal logics of RCC relations are undecidable over the real line [7]. The MSO encoding avoids these barriers by reducing each satisfiability instance to a single sentence in the decidable theory of $(\mathbb{R}, <)$.

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